RELATED RATES QUIZ

BLAKE FARMAN

Lafayette College

Name: Solutions

1. The volume of a tree is given by $V = \frac{1}{12\pi}C^2h$, where C is the circumference of the tree in meters at ground level, and h is the height of the tree in meters. Both C and h are functions of time, t, in years.

(a) Find a formula for $\frac{\mathrm{d}V}{\mathrm{d}t}$. What does it represent in practical terms?

$$V' = \frac{dV}{dt} = \frac{d}{dt}\left(\frac{1}{12\pi}C^2h\right) = \frac{1}{12\pi}\frac{d}{dt}(C^2h)$$

$$= \frac{1}{12\pi}\left[\frac{d}{dt}(C^2)h + C^2\frac{d}{dt}h\right] = \boxed{\frac{1}{12\pi}\left[2CC'h + C^2h'\right]}$$

This represents the rate at which the volume of the tree changes over time.

(b) Suppose the circumerence grows at a rate of 1 meter every 5 years, and the height grows at a rate of 4 meters per year. How fast is the volume of the tree growing when the circumference is 5 meters and the height is 22 meters?

Given: $C' = 1/5$ meters/year, $h' = 4$ meters/year, $C = 5$, $h = 22$

$$V' = \frac{1}{12\pi}\left[2(5)(\tfrac{1}{5})(22) + (5)^2(4)\right] = \frac{1}{12\pi}\left[44 + 100\right]$$

$$= \frac{144}{12\pi} = \frac{12^2}{12\pi} = \boxed{\frac{12}{\pi}\ m^3/year}$$